ENG 101

<u>Lesson - 9</u>

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[1]Computers are machines designed to process, electronically, specially prepared pieces of information which are termed data. Handling or manipulating the information that has been given to the computer, in such ways as doing calculations, adding information or making comparisons is called **processing.** Computers are made up of millions of electronic devices capable of sorting data or moving them, at enormous speeds, through complex circuits with different functions.
[2]All computers have several characteristics in common, regardless of make or design. Information, in the form of instructions and data, is given to the machine, after which the machine acts on it. And a result is then returned. The information presented to the machine is the **input;** the internal manipulative operations, the **processing;** and the result, the output. These three basic concepts of input, processing, and output occur in almost every aspect of human life whether at work or at play.
For example, in clothing manufacturing, the input is the pieces of cut cloth, the processing is the sewing together of these pieces, and the output is the finished garment.

INPUT → COMPUTER → OUTPUT

↕

SEC. STORAGE

[3]Figure 3.1 shows schematically the fundamental hardware components in a computer system. The centerpiece is called either the computer, the processor, or, usually, the **central processing unit** (**CPU**). The term 'computer' includes those parts of **hardware** in which calculations and other data manipulations are performed, and the high-speed internal memory in which data and calculations are stored during actual execution of programs. Attached to the CPU are the various **peripheral devices** such as card readers and **keyboards** (two common examples of input devices). When data or programs need to be saved for long periods of time, they are stored on various **secondary memory** devices or **storage devices** such as magnetic tapes or magnetic disks.
[4]Computers have often been thought of as extremely large adding machines, but this is a very narrow view of their function. Although a computer can only respond to a certain number of instructions, it is not a **single-purpose** machines since these instructions can be combined in an infinite number of sequences. Therefore, a computer has no known limit on the kinds of things it can do; its versatility is limited only to the imagination of those using it.
[5]In the late 1950s and early 1960s when electronic computers of the kind in use today were being developed, they were very expensive to own and run.
Moreover, their size and reliability were such that a large number of support personnel were needed to keep the equipment operating. This has all changed now that computing power has become portable, more compact, and cheaper.
[6]In only a very short period of time, computers have greatly changed the way in which many kinds of work are performed. Computer can remove many of the routine and boring tasks from our lives, thereby leaving us with more time for interesting, creative work. It goes without saying that computers have created whole new areas of work that did not exist before their development.

Exercises1Main Idea Which statement or statements best express the main idea of the text? **Why did you eliminate the other choices?**

☐ 1.Computers have changed the way in which we live.

☐ 2.All computers have an input, a processor, an output and a storage device.

☐ 3.Computers have decreased man’s workload.

☐ 4.All computers have the same basic hardware components.

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2 Understanding the passage:

Decide whether the following statements are true or false (T/F) by referring to the information in the text. Then, make the necessary changes so that the false statements become true.

T	F	
☐	☐	1. All information to be processed must be prepared in such a way that the computer will understand it.
☐	☐	2. Because of the complex electronic circuitry of a computer, data can be either stored or moved about at high speeds.
☐	☐	3. Not all computers can process data given to them and produce results.
☐	☐	4. The basic concepts of data processing are restricted to computers alone.
☐	☐	5. The processor is the central component of a computer system.
☐	☐	6. All other devices used in a computer system are attached to the CPU.
☐	☐	7. Memory devices are used for storing information.
☐	☐	8. Computers are very much restricted in what they can do.
☐	☐	9. Computers today cost less, are smaller, and need fewer people to operate them than in the past.
☐	☐	10. Computer haven’t changed our working conditions very much.

3 Locating Information:

Find the passages in the text where the following ideas are expressed. Give the paragraph reference.

………. 1. All computers are basically the same.

………. 2. Then arithmetic and / or decision-making operations are performed.

………. 3. Computer are limited by man’s imagination more than anything else.

………. 4. All the equipment used in a computer system is the hardware.
………. 5. Computers are electronic machines used for processing data.
………. 6. If programs or data need to be kept for a long time, they are stored on tapes or disks.
………. 7. First the computer accepts data.
……….8. Finally, new information is presented to the user.

4 Contextual reference:

Look back at the text and find out what the words in **bold** typeface refer to.

1. **Which** are termed data (p.1)…………
2. Or moving **them** (p.1)…………
3. The machine acts on **it** (p.2)…………
4. **They** are stored on (p.3)………
5. **It** is not a single-purpose machine (p.4)………
6. The kinds of things **it** can do (p.4)………
7. Of those using **it** (p.4)………
8. **They** were very expensive to own (p.5)………
9. Moreover, **their** size and reliability (p.5)………
10. **That** did not exist (p.6)………

5 Understanding words:

Refer back to the text and find synonyms (i.e. words with a similar meaning) for the following words.

1. called (p.1)……………………
2. tremendous (p.1)……………………
3. ideas (p.2)……………………
4. react (p.4)……………………
5. take away (p.6)……………………

Now refer back to the text and find antonyms (i.e. words with an opposite meaning) for the following words.

6. taken away (p.2)……………………
7. wide (p.4)……………………
8. limited (p.4)……………………
9. immovable (p.5)……………………
10. after (p.6)……………………

6 Word forms:

First choose the appropriate form of the words to complete the sentences. Then check the difference of meaning in your dictionary.

1. Imagination, imagine, imaginable, imaginative, imaginary
a. A computer is limited in its ability by the ……………. of man.
b. Some people are good at inventing …..…… stories.
c. It is practically impossible to …………… the speed at which a computer calculates numbers.
2. Addition, add, added, additional, additional, additive
a. Many terminals can be …………… to a basic system if the need arises.
b. …………….. And subtraction are two basic mathematical operations.
c. when buying a system there is often no …………… charge for the programs.
3. Complication, complicate, complicated, complicating, complicatedly
a. There can be many…………… involved in setting up a computer in an old building.
b. It is sometimes a very ………… process getting into a computer installation for security reasons.
c. It is sometimes very …………….. to explain computer concepts.
4. Difference, differ, different, differently, differential, differentiate

a. There isn't a very big …..…………. in flowcharting for a program to be written in Cobol or Fortran.b. There are many …………….. computer manufactures today, and a buyer must be able to ……………..between the advantages and disadvantages of each.
c. The opinions of programmers as to the best way of solving a problem often……….. greatly.

5.Reliably, rely on, reliable, reliability

a. computers are …………………… machines.

b. If you don't know the meaning of a computer term, you cannot always………………… an all-purpose dictionary for the answer.

c. Computers can do mathematical operations quickly and………………… .